

Matrix inequalities and norms

Modulus-order determinant comparison with an arbitrary base power

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Rating rationale: Arbitrary base powers and indefinite Hermitian factors make extension of the squared-base theorem challenging; the comparison has specialist importance for determinant inequalities.

Resolution — 2026-09-11

Negative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review:** **PASS**.

A rational positive definite A and invertible indefinite Hermitian B in dimension three, with $k = 6$ and $p = 8$, reverse the proposed determinant comparison. The exact right-minus-left gap is $21036678407451/156250000000000 > 0$. The known $k = 2$ theorem and the variant $B > 0$ are not contradicted.

The exact target is resolved. The complete negative answer now has [Lean proof and verification evidence](#). The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; the 2026-09-11 review was independent agent verification, without external human peer review or formal certification. The later Lean verification below covers the complete negative answer. [Submission record](#).

Lean proof and verification evidence — 2026-09-12

Mathematical counterexample: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Lean formalization:** George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, with AI-agent assistance.

The [proof at revision c0c5ece](#) establishes the complete negative answer with the actual continuous functional calculus and matrix modulus. Its exact admissible dimension-three witness at $k = 6, p = 8$ has right-minus-left determinant gap $21036678407451/156250000000000 > 0$. The final negation retains every positive dimension, every complex positive definite A , every invertible Hermitian B , and all non-negative real exponents. Every witness hypothesis is proved; positivity or commutation of B is not added.

[Solution.lean](#) exports:

- `NLA.MI29.spectralPower_natCast`: actual spectral powers agree with matrix natural powers, including exponent zero.
- `NLA.MI29.modulus_power_eight`: the genuine square-root modulus, its positivity and the eighth-power reduction.
- `NLA.MI29.comparison_positive_real`: both determinants are positive real numbers under the original hypotheses.
- `NLA.MI29.counterexample`: the full admissible witness, analytic reductions, exact determinants and strict violation.
- `NLA.MI29.not_modulusDeterminantConjecture`: negation of the complete original conjecture.

Two independent agents [reviewed the frozen statements and complete proof](#) against the original target. [Linux run 34706412510](#) executed the actual sandboxed Comparator on GitHub Actions Ubuntu 24.04, matched all five declarations and replayed the solution through Lean’s default kernel. The [archived logs and independent operational audit](#) retain the immutable source hashes, original artifact digests and successful sandbox, raw-kernel and axiom-rejection controls. The [transitive axiom report](#) contains only *propext*, *Classical.choice* and *Quot.sound*. This catalog reviewed the remote Linux execution; local macOS proof reviews are recorded separately. These are independent AI-agent reviews, not external human peer review or source-author endorsement. Comparator checks formal identity and kernel trust; the statement referees separately reviewed correspondence with the original prose.

The proof pins [Lean 4.33.1](#), [LeanCert 621a43d](#) and [Mathlib 0df444a](#), with [all dependency revisions locked](#). Generic CFC identities reduce the calculation to repeated squaring and exact three-dimensional determinants; one kernel-mode LeanCert rational point certificate supplies the strict scalar gap. The [formalization manifest](#), [numerical targets](#) and [project guide](#) record scope and reproduction. The source’s separate singular-input extension, established $k = 2$ result and positive- B variants are outside these exports.

From a checkout of the verified proof revision, on an isolated non-root Linux host meeting the [shared harness prerequisites](#), run:

```
tools/lean/bootstrap.sh /absolute/path/to/nla-lean-tools
tools/lean/selftest.sh /absolute/path/to/nla-lean-tools
tools/lean/verify.sh \
  matrix-inequalities-and-norms/MI-29/lean \
  /absolute/path/to/nla-lean-tools
```

Problem statement

Does the inequality

$$\det(A^k + |AB|^p) \geq \det(A^k + |BA|^p)$$

hold for every integer $n \geq 1$, every positive definite $A \in \mathbb{C}^{n \times n}$, every invertible Hermitian $B \in \mathbb{C}^{n \times n}$, and every pair of real parameters $k, p \geq 0$? Here $|X| = (X^*X)^{1/2}$, all powers of positive definite matrices use spectral functional calculus, and $X^0 = I$.

Both determinants are positive real numbers. The same conjecture is stated for semidefinite A and Hermitian B in the source; invertible inputs avoid zeroth-power conventions, while the positive-exponent singular cases follow by continuous approximation. No commutation between A and B is assumed.

The two moduli have the same singular spectrum, but adding A^k tests their alignment with the eigenvectors of the base matrix. The question therefore concerns how spectral data and eigenvector geometry interact in determinant estimates for matrix functions.

References

1. M. M. Ghabries, *Contributions to Matrix Inequalities and Some Applications*, PhD thesis, University of Angers and Lebanese University (2022), §2.5 and final Open Problems, Problem 2, pp. 109–110. [Thesis uploaded by its author.](#)
2. M. M. Ghabries, H. Abbas and B. Mourad, *On some open questions concerning determinantal inequalities*, *Linear Algebra Appl.* 596 (2020), 169–183, resolution of the earlier $k = 2$ question. [Publisher.](#)
3. M. M. Ghabries, *A log-majorization inequality for normal matrices with applications to determinantal inequalities and geometric means*, arXiv:2607.21163v1 (2026), §3. [Full text.](#)

Status check (2026-09-10): the thesis states the full question after proving $k = 2$ for all $p \geq 0$, and further cases when both matrices are positive semidefinite. The July 2026 paper extends the known squared-base result to normal matrices, but does not claim arbitrary k here. Searches for the exact titles, “modulus determinant arbitrary power conjecture Ghabries”, and 2025/2026 found no complete resolution. The negative-power companion in the thesis is not counted separately.